ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5,MODULE IV

COMPUTERIZED DATABASE

CAT2

UNIT CODE: 0612 451 08A

Total Marks: 30
Time: 1 hour
Instructions:

- ✓ *Answer **ALL** questions.*
- ✓ *Write SQL statements neatly with correct syntax.*
- ✓ *Write you name and other details in the spaces provided*
- ✓ *Responses to questions should be given in English only*

Candidate name……..admno……

Date……sign……

Question 1 (2 marks)
Define a **NULL** value in a database.

Question 2 (2 marks)
State the SQL command used for each of the following operations: adding, retrieving, modifying, and removing records.

Question 3 (2 marks)
Define a **many-to-one relationship** and give one example.

Question 4 (2 marks)
Distinguish between **primary indexing** and **secondary indexing**.

Question 5 (2 marks)
Differentiate between **entity integrity** and **referential integrity**.

Question 6 (4 marks)
For each of the following relationships, give **one** real-life example and explain how it is implemented using keys:

a) One-to-one (2 marks)

b) One-to-many (2 marks)

Question 7 (2 marks)
Explain why a primary key column cannot contain NULL values.

Question 8 (4 marks)
Explain what is meant by **referential integrity**, and describe **two** actions a database can take when a parent record that is referenced by child records is deleted.

Question 9 (2 marks)
State what SQL stands for and give **one** reason why it is widely used.

Question 10 (2 marks)
Explain how a **many-to-many relationship** is implemented in a relational database.

Question 11 (2 marks)
Consider the table **Students** with the columns student_id (primary key), first_name, last_name, course, and gpa.

Write an SQL statement to insert a student with ID 101, first name "Amina", last name "Otieno", course "IT", and GPA 3.6.

Question 12 (2 marks)
Using the **Students** table in Question 11, write an SQL statement to display the first name and course of all students whose GPA is above 3.0.

Question 13 (2 marks)
Using the **Students** table in Question 11, write an SQL statement to change the course of the student with ID 101 to "Business".

END OF CAT